MILITARY SPECIFICATION

TEST SET, DIGITAL CARD, TS-3349/APM

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall provide signals and measuring devices necessary for the checkout of the digital portion of the RT-1015/APN-194(V) Radar Altimeter when used in conjunction with the Epsco Test Adapter, 2019 and the APM-347 or TS-1979A Altimeter Test Sets.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for
MIL-T-18303	Test Procedure; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration; in so doing the following shall govern:

(a) Parts and materials requirements shall be in accordance with MIL-T-21200. Parts selected for use which are considered nonstandard per MIL-T-21200 shall be selected to meet the design requirements. Such parts shall only be used when a standard MIL equivalent part is not available or when delivery time of the standard part is beyond the time limit required to meet the contract delivery schedule.

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(b) Semiconductor Devices - Semiconductors shall be chosen in accordance with MIL-STD-454, Requirement 30.

3.1.1 Nonstandard Parts and Material Approval - Approval for use of nonstandard parts and materials shall be obtained in accordance with MIL-T-21200.

3.2 Design and Construction - Design and construction of the equipment shall be in accordance with MIL-T-21200 for Class 3 equipment.

3.2.1. Total Weight - The total weight of the equipment including cable, shall be a minimum consistent with good design and shall not exceed 20 pounds.

3.2.2 Reliability

3.2.2.1 Operating Life - The equipment shall have a minimum total operating life of 45,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.2 Reliability in Meantime Between Failure - The calculated mean (operating) time between failure shall be 5,000 hours.

3.2.3 Cabling and Connections - The equipment shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.2.3.1 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with MIL-W-5088. The external wiring may be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. Interference control requirements of MIL-T-21200.

3.2.3.1.1 Wires - Individual wires shall have white jackets. Wire identification shall not be a requirement.

3.2.4 Control Panels - All rack mounted control panels shall conform to the applicable requirements as specified in MIL-T-21200.

3.2.5 Interchangeability - The equipment shall meet the applicable interchangeability requirements as specified in MIL-T-21200.

3.2.6 Interference Control - The equipment shall be designed to meet the radio interference control requirements of MIL-T-21200 minimizing the generation of Radio Interference.

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3.2.7 Maintenance Provisions and Field Testing - Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Identification - Nomenclature assignment and identification approval for equipment identification shall be in accordance with MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 6.0 VAC, 400 ± 20 Hz.

3.2.10 Service Conditions - The equipment shall operate satisfactorily when subjected to the environmental service conditions in accordance with MIL-T-21200 for Class 3 equipment.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 10 minutes at extreme service conditions.

3.2.12 Primary Input Power - The equipment shall give the specified performance from the following power source with characteristics as specified in MIL-T-21200. The power required shall not exceed:

(a) 115 ± 6.0 VAC, 400 ± 20 Hz, Single Phase, 100 VA.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The Digital Test Adapter shall provide for test of the RT-1015/APN-194(V) Altimeter Digital portion at the bench or hangar level.

3.3.1.1 Test capabilities shall be provided as follows:

(a) Display the parity bit output from the APN-194 altimeter.

(b) Display the control bit.

(c) Display the validity bit.

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(d) Decode and display the digital altitude output from the APN-141(V) Altimeter.

(e) Provide read and shift pulses to the APN-194(V) Altimeter system.

3.4 Detail Requirements

3.4.1 The Digital Test Adapter shall meet the following requirements:

3.4.1.1 Digital inputs and outputs to and from the RT-1015/APN-194 Radar Altimeter - The Digital Test Adapter shall generate the necessary digital commands to the unit under test to enable the APN-194(V) altimeter to transfer altitude, validity, parity and control information back to the Digital Test Adapter. The Adapter will then display the state of the control bit, validity bit and the parity bit and decode and display the altitude information in decimal form. The data word from the AN/APN-194(V) altimeter is in the following format. The word is in serial form and the control bit is transferred first.

Bit No.	Designation	Logical Value (True)
1	Control	1
2	Validity	1 = Valid
3	Data LSB	1 = 1.0 feet
4	Data LSB	1 = 2.0 feet
5	Data LSB	1 = 4.0 feet
6	Data LSB	1 = 8.0 feet
7	Data LSB	1 = 16.0 feet
8	Data LSB	1 = 32.0 feet
9	Data LSB	1 = 64.0 feet
10	Data LSB	1 = 128.0 feet
11	Data LSB	1 = 256.0 feet
12	Data LSB	1 = 512.0 feet
13	Data LSB	1 = 1024.0 feet
14	Data LSB	1 = 2048.0 feet
15	Data LSB	1 = 4096.0 feet
16	Spare	0
17	Spare	0
18	Spare	0
19	Spare	0
20	Parity	Odd

The commands to the Altimeter under test are as follows:

(a) Read Pulse = Discrete "1" Read.

(b) Shift Clock-true = Differential 5 KHz to 1 MHz, 50% ± 10 percent duty cycle.

(c) Shift Clock-pulse = Differential 5 KHz to 1 MHz, 50% ± 10 percent duty cycle.

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(d) Signal Ground = Common or chassis ground.

The shift pulse characteristics and timing relationships are shown in Figure 1.

3.4.1.2 Form Factor

Panel Size 10 7/8" long, 6 1/2" wide

Overall Height Including Handles = 7 1/2"

Weight 20 pounds maximum.

3.4.1.3 Controls (Front Panel)

1. Interrogation Rate Control - The Interrogation Rate Control used to control the rate at which the Digital Test Adapter interrogates or asks the RT-1015/APN-194(V) Radar Altimeter to send information to the test adapter. This rate is adjustable from 2 times a second to 20 times a second.

2. Shift Clock Frequency Switch - Selects the frequency of the shift clocks sent to the altimeter under test. The range of clock frequencies shall be 0-5 KHz-125 KHz and 1 MHz

3. Lamp Test - This switch is used to test the segments of the altitude decimal display. Closing the switch will cause all seven segments of each character to illuminate.

3.4.1.4 Controls (Internal)

1. + 5 VDC Power Supply Adjust - Adjusts + 5 VDC output to the proper value.

3.4.1.5 Electrical Connections - Connections to external circuits,

Ref. Des.	Function	Recent Type
P1	Signals and power	MS3116Ek4-15P
J1	Data in "test"	M39012/21-0001
J2	Data in compliment "test"	M39012/21-0001
J3	Read pulse "test"	M39012/21-0001
J4	Shift clock "test"	M39012/21-0001
J5	Shift clock compliment "test"	M39012/21-0001

3.5 Associated Equipment - This equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

(a) Epsco 2019 Test Adapter

(b) APM-347 Test Set or TS-1979A Test Set

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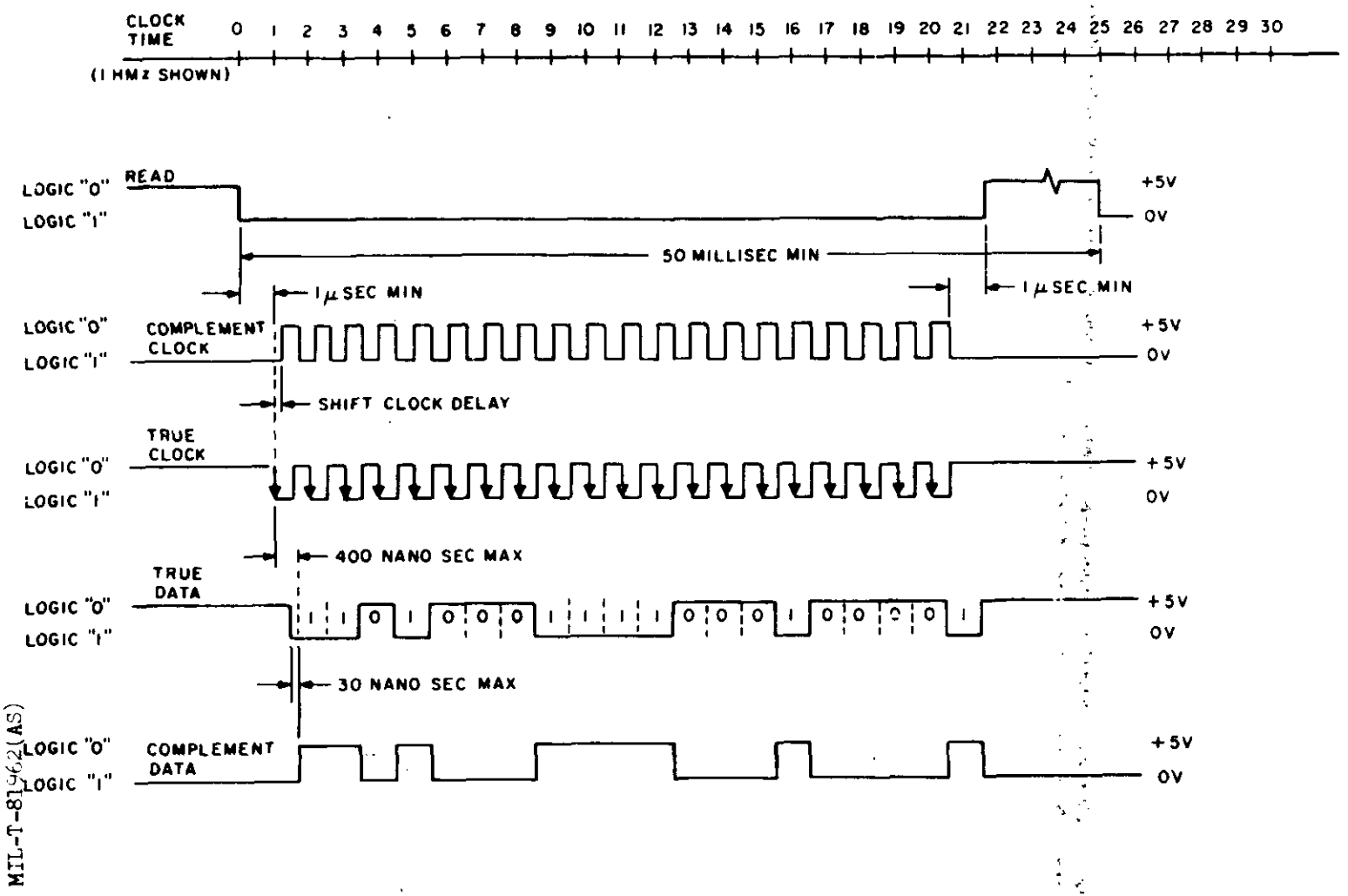


FIGURE 1 DIGITAL TIMING RELATIONSHIPS

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4. QUALITY ASSURANCE

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests
(b) Production Tests
(c) Acceptance Tests

4.1.1.1 Preproduction Tests - When specified in the contract, preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. The tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.1.1.1.1 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Within the limits specified in MIL-T-21200, first article tests shall include environmental tests in accordance with MIL-T-5422 and interference test in accordance with MIL-STD-462 to the MIL-STD-461 requirements.

4.1.1.1.1.1 Preproduction Approval - Approval of the test sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the test sample. Prefabrication of production equipment prior to the approval of the test sample is at the contractor's own risk. The approved test sample may be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

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4.1.1.1.2 Production Equipment - Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.1.1.2 Production Tests - Each production unit shall undergo tests to insure compliance with the requirements of this specification. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved test sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.1.1.2.1 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.1.1.2.2 Reconditioning of Production Test Sample - On completion of the production test, the equipment may be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor may resubmit the equipment for acceptance.

4.1.1.3 Acceptance Tests - The supplier shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Suppliers not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The supplier shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the supplier. Acceptance tests shall consist of individual unit tests. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product.

4.1.1.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability specified herein. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational test

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4.1.1.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.1.1.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. Approval of the operational test procedure must be obtained prior to testing.

4.1.1.4 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Test specifications shall be prepared in accordance with MIL-T-18303.

4.1.3 Presubmission Testing - No item, part or complete equipment shall be submitted until it has been previously tested and inspected by the supplier and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.1.4 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the defects found in the original shall be furnished.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract.

6. NOTES

6.1 Intended Use - This equipment is intended to provide the test capability required for fault isolation, repair and adjustment of the AN/APN-194(V) Altimeter when used in conjunction with Epsco Adapter 2019 and the APM-347 or TS-1979A Test Sets down to the module level.

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ± 6.0 VAC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

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6.3 Performance Objectives - Minimum size and weight simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processed other than as specified in MIL-T-21200, it is desired their use be investigated.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This Specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Definition - The following non-military definition shall apply throughout this specification.

6.5.1 Supplier - The facility fabricating, assembling, acceptance testing and furnishing the equipment in accordance with the contract specifications.

Preparing activity:
Navy - AS
(Project No. 4920-N311)

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DOCUMENT IDENTIFIER AND TITLE MIL-T- 81962(AS)
TEST SET DIGITAL CARD TS-3340/AP

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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